

Off-Road Autonomous Challenge

1. The challenge

Design and build an autonomous ground vehicle that completes a rough off-road course. During the run the vehicle collects balls, scores them into a bucket, and finishes by pulling itself up a bar.

2. The course

One continuous run through the following sections, in order.

Slalom — a line of cones or gates the vehicle weaves through without contact. Also can pick up balls to be carried thru the course for more points before the slalom!

Washboard — a run of fixed transverse bumps of varying height. Tests suspension and stability; body angle is measured by IMU (see Scoring).

Hill — an incline and descent.

Ball task — small tennis balls staged on the course; the vehicle collects them and shoots them into a bucket from one of three scoring lines.

Final climb — a horizontal bar at the finish; the vehicle engages the bar and pulls itself up.

3. Scoring

Driving sections (score at the highest tier reached):

Tier (T)	Description	Hill/Washboard	Slalom
T3	Cleared fully, autonomously	15	10
T2	Autonomous but partial (up not down / cone knocked)	10	7
T1	Completed non-autonomously	5	3
T0	Not cleared fully & cleanly (off-track, hit cone, etc)	0	0

Ball task (per ball landed in the bucket):

Line	Points per ball (1.5x for balls carried thru course)
Line 1 (closest)	3 (4.5)
Line 2 (middle)	5 (7.5)
Line 3 (farthest)	12 (18)

Autonomous collection and shot: full points

Manual assist: $\frac{1}{3}$

Missed bucket: 0

Climb:

Tier	Result	Points
T3	Fully raised	20
T2	Off the ground	10
T1	Hooked, but no lift	6
T0	No hook	0

Autonomous: full points

Assisted: ½ points

Stability in washboard zone:

Stability scoring is calculated as follows–

$$\text{tilt} = \sqrt{\text{roll}^2 + \text{pitch}^2}$$

Average the squares of the tilt value for each time step the IMU measures

$$(\text{tilt}_1^2 + \text{tilt}_2^2 + \text{tilt}_3^2 + \text{tilt}_4^2 + \dots + \text{tilt}_N^2) / N = \text{avg_of_squares}$$

$$\text{preliminary_tilt_score} = \sqrt{\text{avg_of_squares}}$$

$$\text{final_stability_score} = (-1) * \text{preliminary_tilt_score} * 2.5$$

The final_stability_score is subtracted from your points total (signified by the “-1” above)

(no, your IMU may not have its own gyro/suspension etc)

Speed (time bonus):

Finish Time	Bonus
Finish under 1:00	+50
Finish under 1:45	+15
No finish or >1:45	0

4. Run format

3 attempts per ‘competition’ – best score counts.

- If you do not finish, you score for the sections completed
- On "go," no contact with the robot
- Any other contact ends the run; the score stands from that point
- The run ends at the completed final climb, time expiration, or a judge's stop
- No restarts

- Run is a fail if it leaves the washboard or hill at all (other than traveling straight off it on the other side)

5. Safety

1. Approved safety mechanism (kill switch, tether, etc) to be approved by judges
2. Battery secured against ejection if the robot tips (strap, bolt, or bracket), with a fuse or breaker rated for the system.
 - a. Batteries & power systems must be approved by judges before purchase.
3. No exposed or dragging wiring; route and secure all leads.
4. A mentor or judge may stop any run for safety. This is not a penalty; the run may be restarted at their discretion.
5. Safety glasses in the build area at all times.

6. Vehicle size

Footprint (measured by contact patch centers of wheels in each corner): 15 in × 15 in (judges to clarify for designs that don't apply)

Maximum height at start: 8 in (can deploy arms/climbers/etc after starting)

7. Vehicle components

1. Design and fabricate your own chassis and suspension
2. Motors, wheels, and electronics may be COTS
3. An IMU is required, and your code must read pitch & roll during the run

While teams are allowed to use some off-the-shelf parts for your vehicles, a majority of the vehicle components should be custom designed and fabricated.

Parts that teams **should** design and fabricate (as determined by judges):

- Chassis
- Body Parts
- Suspension Arms
- Motor Mounts
- Enclosures for electronics
- Mechanical intakes, arms, grabbers & shooters
- Steering mechanicals

8. Inspection

Passed before your first run. Self-check beforehand:

- Fits in the vehicle size boundaries
- Safety mechanism approved
- Battery secured
- No exposed or loose wiring
- IMU installed and read by the code during runs